

Nature-Inspired AI — Exam (2026-06-22)

reconstructed from transcript, with answer key

*The exam is shown unmarked; correct answers and justifications are in the answer key at the end.
† = question reconstructed from memory (Q1 and Q9 were torn off; the two reconstructed questions fill those slots).*

Exam

1. † (reconstructed) We use GP to evolve a controller for an irrigation drone on a 10×10 grid.
Perceptions (terminals): `soil_moisture(x,y)`, `crop_health(x,y)`, `water_remaining`, `current_x`, `current_y`.
Actions: `irrigate()`, `skip()`, `move_{north,south,east,west}()`.
Fitness = number of cells with `soil_moisture` > 0.6 at the end of the episode.
What is wrong with this setup?
 - a) The fitness should also account for water wasted and healthy crops irrigated, and the primitive set needs conditionals (e.g. **if-then-else**) so the program can branch on the perceptions
 - b) The fitness should account for the number of cells moved, to reward efficient short paths
 - c) It is all fine
 - d) The action set is incomplete: without a `refill()` action the drone cannot replenish water, so no valid solution exists
2. What is the standard recombination operator in GP?
 - a) Uniform crossover
 - b) One-point crossover on bit string
 - c) Exchange subtrees
 - d) Arithmetic recombination
3. In DE survival selection, when does the trial vector u_i replace the target vector x_i ?
 - a) Whenever the trial vector passes the crossover probability threshold
 - b) When a uniform random condition is satisfied
 - c) When $f(u_i) < f(x_i)$, i.e. the trial is worse (for min)
 - d) When $f(u_i) > f(x_i)$, i.e. the trial is better than the target
4. In arithmetic crossover for real-valued parents x_1 and x_2 with $\alpha \in [0, 1]$, the offspring x' is:
 - a) $x' = x_1 + x_2$
 - b) $x' = \alpha \times (x_2 - x_1)$
 - c) $x' = \alpha x_1 + (1 - \alpha) x_2$
 - d) $x' = (x_1 \times x_2) / \alpha$

5. Which is an advantage of EAs over traditional gradient-based optimization methods?
 - a) EAs do not require gradient information and can handle noisy, black-box or non-differentiable objective functions
 - b) EAs are guaranteed to find the global optimum in polynomial time
 - c) EAs are always faster for any problem
 - d) EAs require a small population when the search space is high-dimensional
6. In ACO, pheromone evaporation serves the purpose:
 - a) It is used only in the Rank-Based Ant System
 - b) It replaces the cognitive component from PSO
 - c) It prevents unlimited accumulation of pheromone and allows the algorithm to forget poor solutions over time
 - d) It increases pheromone on the best path found so far
7. Which of the following is NOT a type of terminal?
 - a) Variables (x, y, z)
 - b) Zero-arity functions (rand, go-left)
 - c) Conditional operators (if-then-else)
 - d) Constant values $(0, 3.5)$
8. NES differs from CMA-ES and Simple ES because it:
 - a) Uses all sampled solutions, including poor ones, to update the probability distribution via the natural gradient
 - b) Replaces mutation with crossover
 - c) Only keeps the best solution
 - d) Does not use a probability distribution to sample solutions
9. [†] **(reconstructed)** We use GP to evolve a sorting algorithm.
Function set: `seq(a,b)`, `swap(i,j)`, `if_greater(i,j,a,b)`, `repeat(k,a)`.
Terminal set: $\{i, j, 0, 1, n-1, +, -, \text{rand}()\}$.
 What is wrong with this setup?
 - a) `rand()` cannot be there: it makes the fitness evaluation non-deterministic
 - b) `rand()` cannot be there (non-deterministic fitness), *and* `+`, `-` cannot be in the terminal set because they are not zero-arity (they require arguments)
 - c) It is fine because it satisfies the sufficiency property
 - d) The terminal set is too small to satisfy the sufficiency property, so no correct sorting program can be expressed
10. Which population initialization will always fill trees with valid functions until the maximum depth?
 - a) Ramped half-and-half
 - b) Random
 - c) Full
 - d) Grow
11. Which of the following describes the role of temperature in SA?

- a) Determines the number of neighbors evaluated
 - b) Defines the size of the initial population
 - c) Sets the maximum number of iterations
 - d) Controls the probability of accepting a worse solution, enabling exploration when high and exploitation when low
12. Apply OX to the parents, keeping positions 2 to 5 (0-indexed) from x_1 :
- $$x_1 = [2, 4, 7, 1, 6, 3, 8, 5, 9], \quad x_2 = [5, 3, 1, 9, 6, 2, 4, 8, 7]$$
- a) [9, 2, 7, 1, 6, 3, 4, 8, 5]
 - b) [5, 9, 7, 1, 6, 3, 4, 2, 9]
 - c) [2, 4, 7, 1, 6, 3, 8, 5, 9]
 - d) [9, 2, 7, 1, 6, 3, 8, 4, 5]
13. Using the best individual as the base vector in DE:
- a) Eliminates the need for crossover
 - b) Increases selective pressure and decreases diversity
 - c) Increases diversity, decreases selective pressure
 - d) Has no effect on selection or diversity
14. In GP, the result classifies the training data perfectly but fails on unseen data. Explanation and remedy:
- a) Population too small; increase it
 - b) The program has overfit; use a validation set and penalize complexity
 - c) Mutation rate too high; reduce it
 - d) Crossover rate too low; increase it
15. In $(\mu + \lambda)$ -ES, which individuals are considered in selection?
- a) Only the λ offspring
 - b) The μ parents
 - c) A random subset of λ and μ
 - d) Both μ and λ
16. In PSO, the cognitive velocity is:
- a) $k \times (x_{\text{best}} - x^{(j-1)})$
 - b) $k \times v^{(j-1)}$
 - c) $k \times (x_{\text{global-best}} - x^{(j-1)})$
 - d) $k \times (x^{(j)} - x_{\text{best}})$

Answer Key

Every *circled* answer on the sheet (Q2–Q16) matches the key. Q1 and Q9 are reconstructed; their keys follow the same logic the course teaches. ✓ marks the correct option below.

- Q1[†]** ✓ **a.** Two real flaws. (1) The fitness only counts wet cells, ignoring the true objective — it never penalizes wasted water nor rewards `crop_health`, so a drone that floods everything scores well. (2) The primitive set has no functions/conditionals, so an evolved tree cannot branch on the perceptions (`soil_moisture`, `water_remaining`, ...) — the sensors are unusable and the set fails sufficiency. (b) path efficiency is a minor concern, not the core flaw.
- Q2** ✓ **c.** Subtree exchange is the canonical GP crossover.
- Q3** ✓ **d.** DE keeps the better of {target, trial} (slide: “choose the better solution”). The item’s inequalities and “better/worse” labels are inconsistent on min vs. max; **d** is the intended “trial is better”.
- Q4** ✓ **c.** Whole-arithmetic crossover = convex combination.
- Q5** ✓ **a.** b/c/d are false absolutes (no guarantees, no free lunch).
- Q6** ✓ **c.** Evaporation = decay preventing stagnation; (d) describes deposit.
- Q7** ✓ **c.** `if-then-else` takes arguments, so it is a function (internal node), not a terminal.
- Q8** ✓ **a.** NES = natural-gradient update on the search distribution using all samples.
- Q9[†]** ✓ **b.** Decisive flaw: + and - are binary (arity 2), so they are *functions*, not terminals — a leaf cannot take arguments. Plus the same `rand()` non-determinism issue. (c) is wrong because satisfying sufficiency does not excuse an invalid terminal/function split; (d) is wrong because the primitives (`swap`, `if_greater`, `repeat`, indices) can express a sorting program, so sufficiency does hold.
- Q10** ✓ **c.** Full uses only functions until the depth limit, then terminals; Grow may stop early.
- Q11** ✓ **d.** High $T \rightarrow$ explore (accept worse), low $T \rightarrow$ exploit.
- Q12** ✓ **a.** Working:

Q12 — OX working

- Copy the block.** 0-indexed positions 2–5 of x_1 are [7, 1, 6, 3], placed at the same slots: $\text{child} = [-, -, \mathbf{7}, \mathbf{1}, \mathbf{6}, \mathbf{3}, -, -, -]$.
- Fill from x_2 ,** starting just after the block (position 7), wrapping, skipping values already present {7, 1, 6, 3}:
 x_2 from pos 7: 4, 8, ~~7~~, 5, ~~3~~, ~~1~~, 9, ~~6~~, 2 $\Rightarrow$ 4, 8, 5, 9, 2.
- Drop into the empty slots** (7,8,9 then wrap to 1,2):
pos7= 4, pos8= 8, pos9= 5, pos1= 9, pos2= 2.
 $\Rightarrow$ [9, 2, 7, 1, 6, 3, 4, 8, 5] = option **a**.
(Option d swaps the 4 and 8 — the trap if you fill x_2 from its start instead of from after the cut.)

- Q13** ✓ **b.** Confirmed by DE slide: “ $v_i = x_{\text{best}} + \dots \rightarrow$ increase selective pressure, decrease diversity.”
- Q14** ✓ **b.** Perfect-train / poor-test is the signature of overfitting; remedy with validation + complexity penalty.
- Q15** ✓ **d.** “+” means parents and offspring compete together (elitist); “,” would be offspring only.

Q16 ✓ **a.** Cognitive term pulls toward the particle's own personal best; (c) is the social term, (b) is inertia.